S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 7

Aim: To apply Support Vector Machine for classification and analyze the impact of different kernels on model accuracy.

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Aim: To apply Support Vector Machine for classification and analyze the impact of different kernels on model accuracy.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

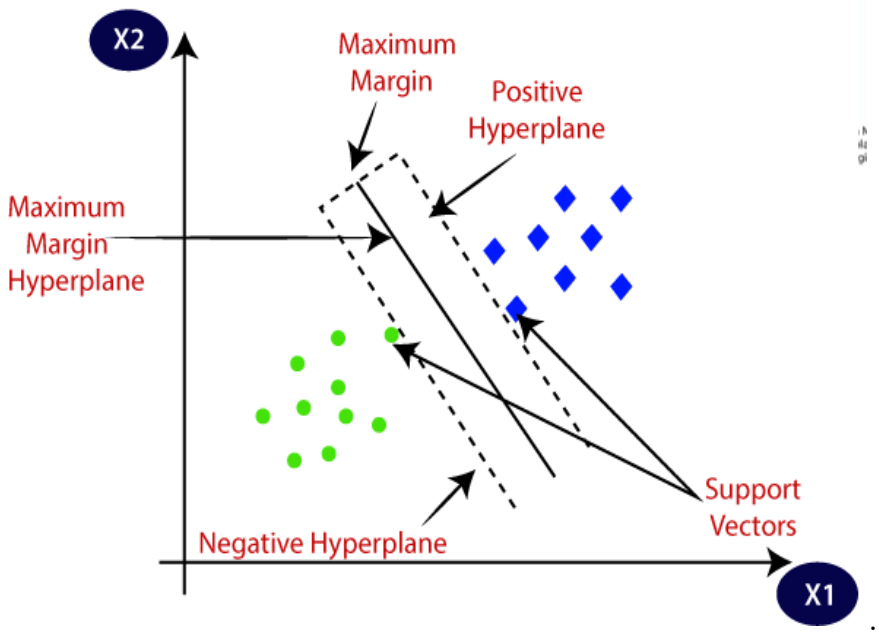
- The objective of the SVM algorithm is to find a hyperplane in an N-dimensional space that distinctly classifies the data points.
- The dimension of the hyperplane depends upon the number of features.
- If the number of input features is two, then the hyperplane is just a line.
- If the number of input features is three, then the hyperplane becomes a 2-D plane. It becomes difficult to imagine when the number of features exceeds three.

THEORY:

Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. However, primarily, it is used for Classification problems in Machine Learning.

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane.

SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:



Types of SVM

SVM can be of two types:

- **Linear SVM:** Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.
- **Non-linear SVM:** Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Hyperplane and Support Vectors in the SVM algorithm:

Hyperplane: There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but we need to find out the best decision boundary that helps to classify the data points. This best boundary is known as the hyperplane of SVM.

The dimensions of the hyperplane depend on the features present in the dataset, which means if there are 2 features (as shown in image), then hyperplane will be a straight line. And if there are 3 features, then hyperplane will be a 2-dimension plane.

We always create a hyperplane that has a maximum margin, which means the maximum distance between the data points.

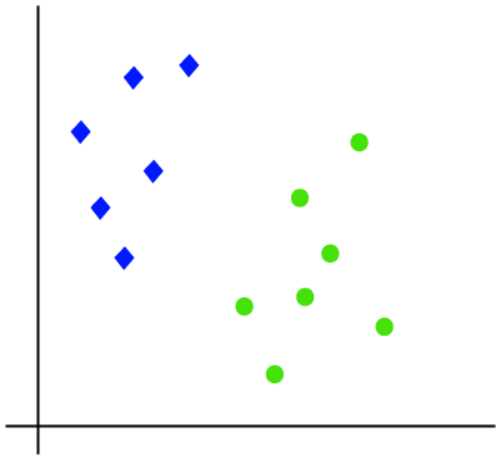
Support Vectors:

The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector. Since these vectors support the hyperplane, hence called a Support vector.

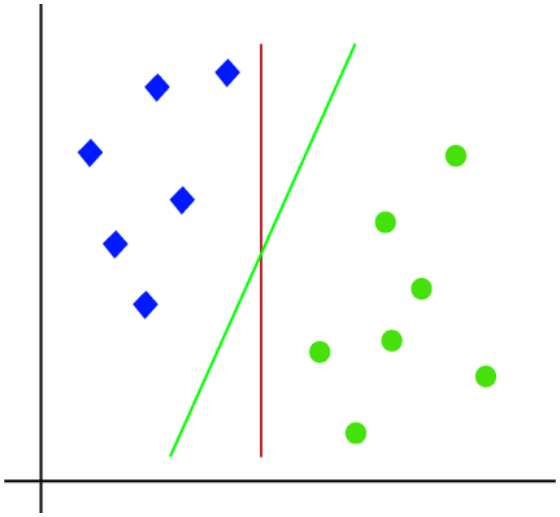
How does SVM works?

Linear SVM:

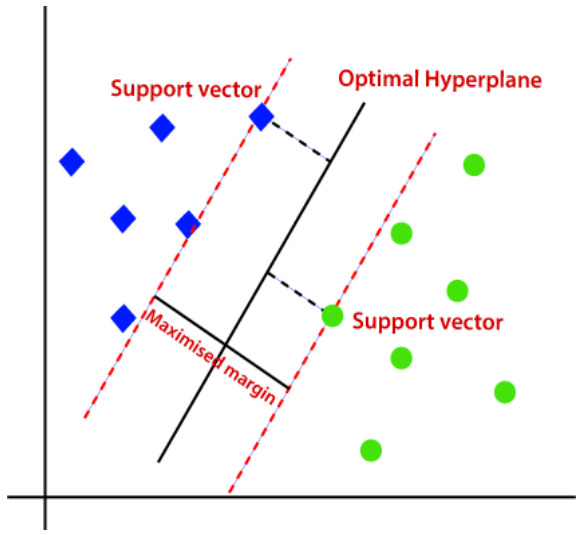
The working of the SVM algorithm can be understood by using an example. Suppose we have a dataset that has two tags (green and blue), and the dataset has two features x1 and x2. We want a classifier that can classify the pair(x1, x2) of coordinates in either green or blue. Consider the below image:



So as it is 2-d space so by just using a straight line, we can easily separate these two classes. But there can be multiple lines that can separate these classes. Consider the below image:

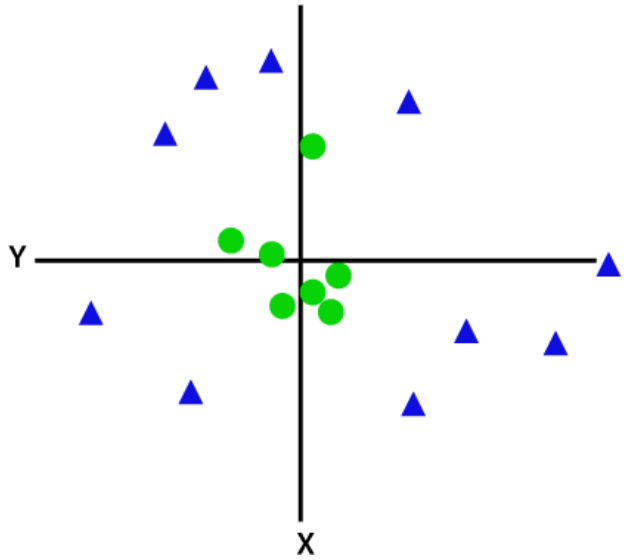


Hence, the SVM algorithm helps to find the best line or decision boundary; this best boundary or region is called as a **hyperplane**. SVM algorithm finds the closest point of the lines from both the classes. These points are called support vectors. The distance between the vectors and the hyperplane is called as **margin**. And the goal of SVM is to maximize this margin. The **hyperplane** with maximum margin is called the **optimal hyperplane**.



Non-Linear SVM:

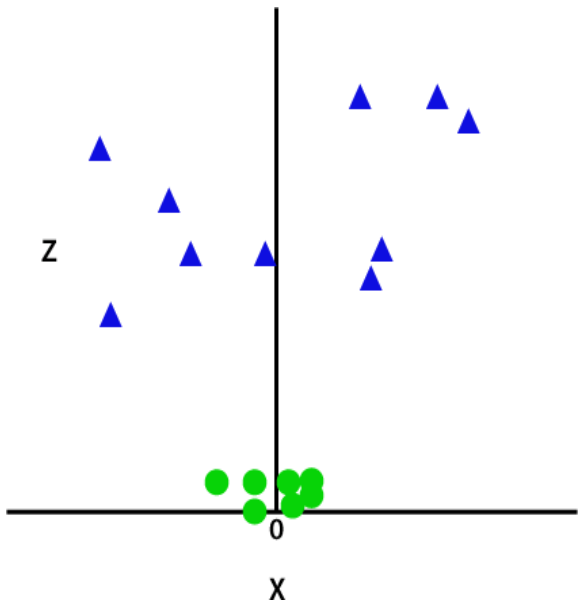
If data is linearly arranged, then we can separate it by using a straight line, but for non-linear data, we cannot draw a single straight line. Consider the below image:



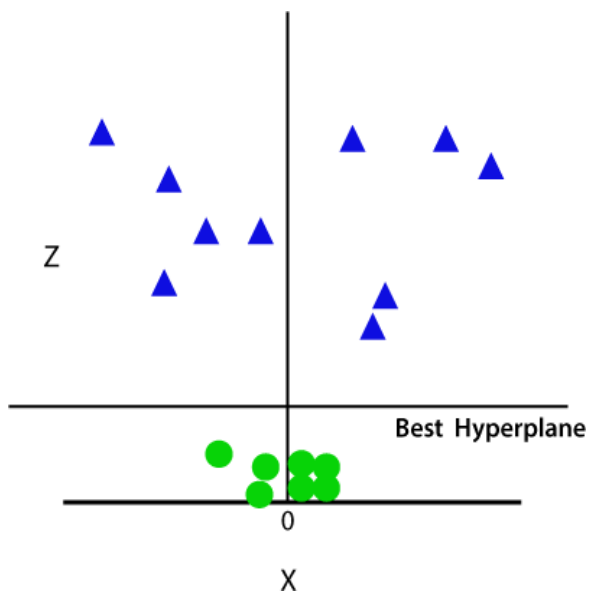
So to separate these data points, we need to add one more dimension. For linear data, we have used two dimensions x and y, so for non-linear data, we will add a third dimension z. It can be calculated as:

$z=x^2+y^2$

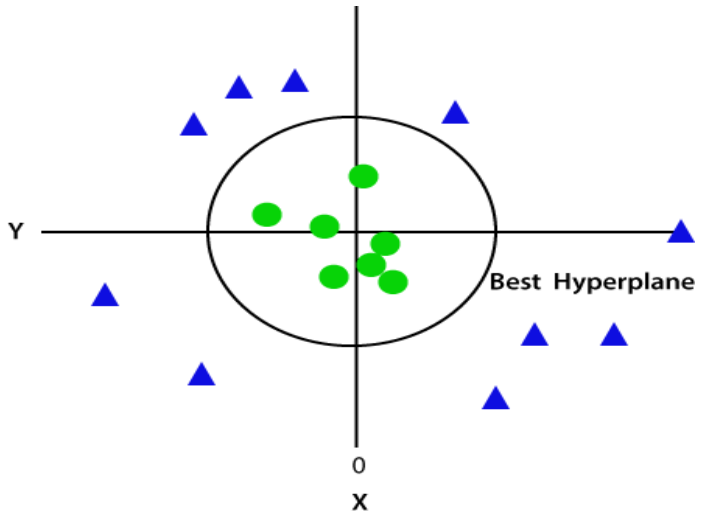
By adding the third dimension, the sample space will become as below image:



So now, SVM will divide the datasets into classes in the following way. Consider the below image:



Since we are in 3-d Space, hence it is looking like a plane parallel to the x-axis. If we convert it in 2d space with z=1, then it will become as:



Hence we get a circumference of radius 1 in case of non-linear data.

Advantages of SVM

1. SVM works better when the data is Linear
2. It is more effective in high dimensions
3. With the help of the kernel trick, we can solve any complex problem
4. SVM is not sensitive to outliers
5. Can help us with Image classification

Disadvantages of SVM

1. Choosing a good kernel is not easy
2. It doesn't show good results on a big dataset
3. The SVM hyperparameters are Cost -C and gamma. It is not that easy to fine-tune these hyper-parameters. It is hard to visualize their impact

PROGRAM CODE:

OUTPUT (SCREENSHOT):

CONCLUSION:

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DISCUSSION AND VIVA VOCE:

1. In a real-world fraud detection system, when would you prefer SVM over Logistic Regression?
2. How does the choice of kernel (linear, RBF, polynomial) impact performance in production systems?
3. What is the role of the regularization parameter C in SVM, and how does it affect overfitting?
4. In high-dimensional datasets like text classification, why is linear SVM often preferred?
5. How would you handle imbalanced data when training an SVM model in a medical diagnosis application?
6. Why is feature scaling mandatory before applying SVM?
7. What are the computational challenges of deploying SVM in large-scale industry datasets?
8. In a real-time prediction system (e.g., credit scoring), what are the limitations of SVM?
9. How do you select hyperparameters (C and gamma) in industry projects?
10. Why might companies prefer tree-based models (Random Forest, XGBoost) over SVM in production?

REFERENCE

- https://www.analyticsvidhya.com/blog/2021/10/support-vector-machinessvm-a-complete-guide-for-beginners/
- https://www.geeksforgeeks.org/support-vector-machine-algorithm/
- https://www.javatpoint.com/machine-learning-support-vector-machine-algorithm